

**EXERCISES TO ACCOMPANY THE SEMINAR ON
FINITE FIELD ARITHMETIC AND ALGEBRAIC GEOMETRY**

- (1) Can a field contain $\mathbb{F}_p$ and $\mathbb{F}_{p'}$ for distinct primes p, p' ?
- (2) Show that a finite integral domain is a field.
- (3) Show that every element of a finite field can be written as a sum of two squares. That is, for each $a \in \mathbb{F}_q$, there exist $x, y \in \mathbb{F}_q$ such that $x^2 + y^2 = a$.
- (4) Let k be any field. Prove that the multiplicative group $k^\times$ is cyclic if and only if k is a finite field.
- (5) (a) Given a mapping $\psi : \mathbb{F}_q \rightarrow \mathbb{F}_q$, there is a unique $f \in \mathbb{F}_q[X]$ of degree at most $q - 1$ such that $\psi(x) = f(x)$ for all $x \in \mathbb{F}_q$.
(b) Conversely, if R is a commutative ring such that every mapping $R \rightarrow R$ is given by a polynomial, then R is a field.
- (6) Let $E \supset F = \mathbb{F}_q$, $N = N_{E/F}$ (norm), $T = T_{E/F}$ (trace). Show for $x \in E$, $T(x) = 0 \leftrightarrow x = y^q - y$ for some $y \in E$, and $N(x) = 1 \leftrightarrow x = y^{q-1}$ for some $y \in E^*$. (These are two instances of “Hilbert’s Theorem 90”.)

Proof. One direction of each is obvious. For the other, assume $T(x) = 0$ and choose any $y \in \overline{E}$ satisfying $x = y^q - y$. Now observe that

$$y^{q^r} = (x + y)^{q^{r-1}} = x^{q^{r-1}} + y^{q^{r-1}} = \cdots = \sum_{i=1}^{r-1} x^{q^i} + y^q = \sum_{i=1}^{i=q} x^{q^i} + x + y = T(x) + y = y.$$

Thus $y \in E$. The same proof strategy works for the norm case. □

- (7) Evaluate the sum and the product of the non-zero elements of $\mathbb{F}_q$.
- (8) Show that $M_n(k)$, the algebra of $n \times n$ matrices over a field k , is a central simple algebra. [*Hint: First prove that the center is the field of diagonal matrices tI where $t \in k$ and I is the identity matrix.*]
- (9) If k is an infinite field and $f(x_1, \dots, x_n)$ is a nonzero polynomial with coefficients in k , show that f is not identically zero on $\mathbb{A}^n(k)$. [*Hint: Try induction on n .*]

Let F be a finite field with $\text{char}(F) \neq 2$. Define the quadratic character $\left(\frac{\cdot}{F}\right)$ over the finite field F by the formula

$$\left(\frac{x}{F}\right) = \begin{cases} 1 & \text{if } x \in F^2 \setminus \{0\} \\ -1 & \text{if } x \notin F^2 \\ 0 & \text{if } x = 0. \end{cases}$$

Here F^2 denotes the set of squares in F ; i.e., $F^2 = \{z^2 : z \in F\}$. The next sequence of problems deals with the quadratic character.

- (1) Show that $\sum_{x \in F} \left(\frac{x}{F}\right) = 0$.
- (2) For fixed $y \in F$, we have

$$\#\{x \in F : x^2 = y\} = 1 + \left(\frac{y}{F}\right).$$

- (3) Fix $a \in F$ and show that

$$\#\{(x, y) : x^2 + y^2 = a\} = q + \sum_{x \in F} \left(\frac{a - x^2}{F}\right).$$

- (4) Show that the restriction of $\left(\frac{\cdot}{F}\right)$ to $F^\times$ is a homomorphism into $\mathbb{C}^\times$: $\left(\frac{xy}{F}\right) = \left(\frac{x}{F}\right) \left(\frac{y}{F}\right)$.
- (5) Show $\left(\frac{xy^{-1}}{F}\right) = \left(\frac{xy}{F}\right)$ for all $y \neq 0$.

- (6) For $E \supset F$, show $\left(\frac{N(x)}{F}\right) = \left(\frac{x}{E}\right)$ for all $x \in E$, where $N = N_{E/F}$ is the norm function. [Hint: Show that the norm map is surjective.]
- (7) For $E \supset F$ and $x \in F$, find the relation between $\left(\frac{x}{F}\right)$ and $\left(\frac{x}{E}\right)$. [Hint: Use the previous exercise.]

For the next two problems, let E be a finite field with $q = p^r$ elements and let $T_E = T_{E/\mathbb{F}_p}$ be the absolute trace.

- (1) Classify all extensions E of $\mathbb{F}_p$ such that the absolute trace vanishes on $\mathbb{F}_p$; i.e., $T_E(x) = 0$ for all $x \in \mathbb{F}_p$.
- (2) Suppose $\psi : E \rightarrow \mathbb{C}^\times$ is an additive character; i.e., $\psi(\alpha + \beta) = \psi(\alpha)\psi(\beta)$ for all $\alpha, \beta \in E$. Show that there exists $\gamma \in E$ such that

$$\psi(\alpha) = \zeta^{T_E(\gamma\alpha)}$$

for all $\alpha \in E$, where $\zeta = e^{2\pi i/p}$.

- (3) Let G be a finite Abelian group and let $G^\vee$ be the character group. Recall that $C^0(G, \mathbb{C})$ is the $\mathbb{C}$ -vector space of functions on G and that we have a pairing on $C^0(G, \mathbb{C})$ given by

$$(f_1, f_2) = \frac{1}{\#G} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Show that any function $f : G \rightarrow \mathbb{C}$ can be written as

$$f = \sum_{\chi \in G^\vee} (f, \chi) \chi.$$

Conclude that the characters of G form a basis for $C^0(G, \mathbb{C})$.

- (4) If G is a finite cyclic group, show that $G \cong G^\vee$.
- (5) In this exercise we want to show that $G \cong (G^\vee)^\vee$ for any finite Abelian group. For each $g \in G$, we have a function $\hat{g} : G^\vee \rightarrow \mathbb{C}^\times$ defined by

$$\chi \mapsto \hat{g}(\chi) = \chi(g).$$

Show that $\hat{g}$ is a character on the group $G^\vee$ and that the map $g \mapsto \hat{g}$ is an isomorphism.

- (6) For each $r \geq 1$, compute the number of points in $\mathbb{P}^3(\mathbb{F}_{p^r})$ on the surface defined by the equation $X_0X_1 - X_2X_3 = 0$. [Answer: $(p^r + 1)^2$]
- (7) Compute the zeta function for the hypersurface in the previous problem. Check that it supports the Weil conjectures (note that it isn't a diagonal hypersurface, so this is a case not proved in the text).